Team 10: Semicolon Squad

Team	Semicolon Squad
Project	Digital Knowledge Platform
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Core features to consider

1. Add failure handling for external service integrations. The system depends on MinIO for storage and Elasticsearch for search, but there is no documented behavior when either service is unavailable. Define whether searches degrade gracefully to database queries and whether uploads queue locally when storage is down.
2. Build a data retention and archival policy. The SDD mentions soft-deletes and audit logging, but there are no rules for how long records are kept, when soft-deleted items are permanently purged, or how much audit log storage to provision.
3. Implement rate-limiting and quota policies for bulk operations. Bulk upload supports 50 files at once, but there is no protection against a single user uploading 50 files of 500 MB each, repeatedly. Add per-user quotas and file-size limits.
4. Add a CI/CD pipeline definition. The deployment diagram shows Docker Compose, but there is no continuous integration setup, automated testing trigger, or secrets management approach documented. Without CI/CD, every deployment is manual and error-prone.

What's missing

- The data model is incomplete. 14 entities are listed in Section 4.1 but only 12 are fully detailed. ArchiveVersion and Fine entities are mentioned in the text but never defined with their constraints and relationships.
- Async job handling is not specified. The architecture sections describe several background tasks, but there is no distinction between synchronous and queued operations, and no failure recovery for stuck jobs.
- No backward compatibility strategy for schema changes. The system will evolve, but there is no plan for how database schema changes are deployed without breaking running instances.
- User onboarding metrics are mentioned but not designed. NFR-DKP024 references "learning curve" but the SDD has no UX strategy for reducing it.
- The link to the repository does not seem to be a working link.
- No API documentation is provided.

Weaknesses of SRS

Section 11 (Risks and Open Questions) contains over 25 genuinely critical unresolved decisions, including cloud versus on-premises hosting, fine calculation rules, LLM provider selection, and vector database choice. These are not risks to monitor; they are blocking decisions that need to be made before implementation starts. Framing them as "to be decided during development" is a recipe for scope creep and mid-sprint pivots.

Weaknesses of SDD

The testing plan in Section 7 is a placeholder, not a plan. There are no test case implementations, no coverage targets, and no resource allocation for manual testing. "Known hard-to-test areas" like barcode scanning are flagged for manual QA, but there is no timeline, no responsible person, and no pass/fail criteria. A testing plan that cannot be executed is not a plan.

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